THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 32.4	$\sum f_i = 50$ and $\sum f_i x_i = 1620$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1620}{50} = 32.4$
2	**(a)** 26	Median class 20–30: $l = 20$, $cf = 16$, $f = 15$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{25-16}{15} \times 10 = 26$
3	**(a)** 77	$l = 70$, $f_1 = 16$, $f_0 = 2$, $f_2 = 10$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{14}{20} \times 10 = 77$
4	**(b)** 6	$\frac{1750 + 35p}{44 + p} = 39.2$ gives $(35 - 39.2)p = -25.2$; $p = 6$
5	**(b)** 11	$91.5 = 90 + \frac{50 - (36 + x)}{20} \times 10$ gives $x = 11$; $x = 11$; then $y = 100 - 68 - x = 21$
6	**(b)** 22	$35 = 30 + \frac{50 - (24 + x)}{10} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 57 - x = 22$
7	**(a)** 30	$\frac{1200}{40}$
8	**(b)** 112	$\sum f_i = 50$ and $\sum f_i x_i = 5600$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5600}{50} = 112$
9	**(d)** 38	Median class 30–40: $l = 30$, $cf = 21$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-21}{5} \times 10 = 38$
10	**(c)** 85	$l = 80$, $f_1 = 13$, $f_0 = 5$, $f_2 = 5$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{8}{16} \times 10 = 85$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2; $\sum f_i = 50$ and $\sum f_i u_i = -13$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-13}{50} = 32.4$ (1).	1+1
12	Cumulative frequencies: 4, 8, 18, 26, 34, 40; $N = 40$, so $\frac{N}{2} = 20$; Median class 50–60: $l = 50$, $cf = 18$, $f = 8$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{20-18}{8} \times 10 = 52.5$ (1).	1+1
13	Highest frequency is 16, so the modal class is 70–80; $l = 70$, $f_1 = 16$, $f_0 = 2$, $f_2 = 10$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{14}{20} \times 10 = 77$ (1).	1+1
14	Take $a = 115$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2; $\sum f_i = 50$ and $\sum f_i u_i = -15$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 115 + 10 \times \frac{-15}{50} = 112$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 50$ and $\sum f_i u_i = -48$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-48}{50} = 25.4$ (1).	1+1+1
16	Cumulative frequencies: 6, 15, 22, 27, 43, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 30–40: $l = 30$, $cf = 22$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-22}{5} \times 10 = 36$ (1).	1+1+1
17	$\sum f_i = 50$ and $\sum f_i x_i = 3360$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{3360}{50} = 67.2$ (1); $l = 70$, $f_1 = 19$, $f_0 = 10$, $f_2 = 3$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{9}{25} \times 10 = 73.6$ (1).	1+1+1
	OR: Take $a = 45$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 50$ and $\sum f_i u_i = -44$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 45 + 10 \times \frac{-44}{50} = 36.2$ (1).	1+1+1
18	Cumulative frequencies: 3, 9, 22, 37, 44, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 110–120: $l = 110$, $cf = 22$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 110 + \frac{25-22}{15} \times 10 = 112$ (1).	1+1+1
	OR: $\sum f_i = 50$ and $\sum f_i x_i = 1960$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1960}{50} = 39.2$ (1); $l = 50$, $f_1 = 14$, $f_0 = 8$, $f_2 = 5$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{6}{15} \times 10 = 54$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i = 45 + p$ and $\sum f_i x_i = 1445 + 35p$ (2); $\frac{1445 + 35p}{45 + p} = 32.4$ gives $(35 - 32.4)p = 13$ (2); $p = 5$ (1).	2+2+1

	OR: Median class is 50–60 with $f = 20$, and the cumulative frequency before it is $cf = 21 + x$ (2); $54 = 50 + \frac{50-(21+x)}{20} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 63 - x = 16$ (1).	2+2+1
20	$\sum f_i = 37 + p$ and $\sum f_i x_i = 4235 + 105p$ (2); $\frac{4235+105p}{37+p} = 112$ gives $(105 - 112)p = -91$ (2); $p = 13$ (1).	2+2+1
	OR: Median class is 120–130 with $f = 20$, and the cumulative frequency before it is $cf = 27 + x$ (2); $120.5 = 120 + \frac{50-(27+x)}{20} \times 10$ gives $x = 22$ (2); $x = 22$; then $y = 100 - 57 - x = 21$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	$\frac{1635+45p}{45+p} = 37.2$ gives $(45-37.2)p = 39$; $p = 5$
2	**(d)** 10	$45 = 40 + \frac{25-(7+x)}{16} \times 10$ gives $x = 10$; $x = 10$; then $y = 50 - 30 - x = 10$
3	**(d)** 16	$54 = 50 + \frac{50-(21+x)}{20} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 63 - x = 16$
4	**(c)** just greater than $\frac{N}{2}$	Median class definition
5	**(d)** 103	$\sum f_i = 100$ and $\sum f_i x_i = 10300$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{10300}{100} = 103$
6	**(c)** 45	Median class 40–50: $l = 40$, $cf = 17$, $f = 6$, $h = 10$; Median $= l + \frac{\frac{N}{2}-cf}{f} \times h = 40 + \frac{20-17}{6} \times 10 = 45$
7	**(c)** 76.4	$l = 70$, $f_1 = 20$, $f_0 = 4$, $f_2 = 11$, $h = 10$; Mode $= l + \frac{f_1-f_0}{2f_1-f_0-f_2} \times h = 70 + \frac{16}{25} \times 10 = 76.4$
8	**(b)** 20	$\frac{5180+45p}{80+p} = 60.8$ gives $(45-60.8)p = -316$; $p = 20$
9	**(b)** 22	$105 = 100 + \frac{50-(20+x)}{16} \times 10$ gives $x = 22$; $x = 22$; then $y = 100 - 65 - x = 13$
10	**(d)** 21	$120.5 = 120 + \frac{50-(27+x)}{20} \times 10$ gives $x = 22$; $x = 22$; then $y = 100 - 57 - x = 21$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cumulative frequencies: 8, 10, 13, 33, 36, 39, 50; $N = 50$, so $\frac{N}{2} = 25$; Median class 40–50: $l = 40$, $cf = 13$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2}-cf}{f} \times h = 40 + \frac{25-13}{20} \times 10 = 46$ (1).	1+1
12	Highest frequency is 13, so the modal class is 80–90; $l = 80$, $f_1 = 13$, $f_0 = 5$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1-f_0}{2f_1-f_0-f_2} \times h = 80 + \frac{8}{16} \times 10 = 85$ (1).	1+1
13	Take $a = 105$, $h = 10$; $u_i = \frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2, 3; $\sum f_i = 100$ and $\sum f_i u_i = -20$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 105 + 10 \times \frac{-20}{100} = 103$ (1).	1+1
14	Cumulative frequencies: 4, 8, 23, 29, 37, 40; $N = 40$, so $\frac{N}{2} = 20$; Median class 60–70: $l = 60$, $cf = 8$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2}-cf}{f} \times h = 60 + \frac{20-8}{15} \times 10 = 68$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take $a = 45$, $h = 10$; $u_i = \frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 40$ and $\sum f_i u_i = -7$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 45 + 10 \times \frac{-7}{40} = 43.25$ (1).	1+1+1
16	Cumulative frequencies: 8, 27, 46, 66, 82, 91, 100; $N = 100$, so $\frac{N}{2} = 50$ (1); Median class 100–110: $l = 100$, $cf = 46$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2}-cf}{f} \times h = 100 + \frac{50-46}{20} \times 10 = 102$ (1).	1+1+1
17	$\sum f_i = 50$ and $\sum f_i x_i = 1860$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1860}{50} = 37.2$ (1); $l = 50$, $f_1 = 14$, $f_0 = 5$, $f_2 = 5$, $h = 10$ (1); Mode: $= l + \frac{f_1-f_0}{2f_1-f_0-f_2} \times h = 50 + \frac{9}{18} \times 10 = 55$ (1).	1+1+1
	OR: Take $a = 75$, $h = 10$; $u_i = \frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 100$ and $\sum f_i u_i = -19$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 75 + 10 \times \frac{-19}{100} = 73.1$ (1).	1+1+1
18	Cumulative frequencies: 10, 13, 16, 23, 43, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 120–130: $l = 120$, $cf = 23$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2}-cf}{f} \times h = 120 + \frac{25-23}{20} \times 10 = 121$ (1).	1+1+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 6080$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{6080}{100} = 60.8$ (1); $l = 70$, $f_1 = 22$, $f_0 = 15$, $f_2 = 19$, $h = 10$ (1); Mode: $= l + \frac{f_1-f_0}{2f_1-f_0-f_2} \times h = 70 + \frac{7}{10} \times 10 = 77$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i = 84 + p$ and $\sum f_i x_i = 8460 + 115p$ (2); $\frac{8460+115p}{84+p} = 103$ gives $(115-103)p = 192$ (2); $p = 16$ (1).	2+2+1

	OR: Median class is 70–80 with $f = 18$, and the cumulative frequency before it is $cf = 20 + x$ (2); $75 = 70 + \frac{50-(20+x)}{18} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 70 - x = 9$ (1).	2+2+1
20	$\sum f_i = 47 + p$ and $\sum f_i x_i = 5415 + 95p$ (2); $\frac{5415+95p}{47+p} = 114$ gives $(95 - 114)p = -57$ (2); $p = 3$ (1).	2+2+1
	OR: Median class is 60–70 with $f = 10$, and the cumulative frequency before it is $cf = 13 + x$ (2); $69 = 60 + \frac{25-(13+x)}{10} \times 10$ gives $x = 3$ (2); $x = 3$; then $y = 50 - 35 - x = 12$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 48	$50 + \frac{-60}{30}$
2	**(a)** 114	$\sum f_i = 50$ and $\sum f_i x_i = 5700$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5700}{50} = 114$
3	**(b)** 73	Median class 70–80: $l = 70$, $cf = 44$, $f = 20$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{50-44}{20} \times 10 = 73$
4	**(c)** 112.5	$l = 110$, $f_1 = 21$, $f_0 = 20$, $f_2 = 18$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 110 + \frac{1}{4} \times 10 = 112.5$
5	**(d)** 15	$\frac{6015 + 85p}{85 + p} = 72.9$ gives $(85 - 72.9)p = 181.5$; $p = 15$
6	**(a)** 18	$67.5 = 60 + \frac{50 - (17 + x)}{20} \times 10$ gives $x = 18$; $x = 18$; then $y = 100 - 76 - x = 6$
7	**(b)** 9	$75 = 70 + \frac{50 - (20 + x)}{18} \times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 70 - x = 9$
8	**(b)** increases by 5	Mean shifts with the data
9	**(b)** 55.6	$\sum f_i = 100$ and $\sum f_i x_i = 5560$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5560}{100} = 55.6$
10	**(d)** 34	Median class 30–40: $l = 30$, $cf = 42$, $f = 20$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{50-42}{20} \times 10 = 34$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Highest frequency is 20, so the modal class is 70–80; $l = 70$, $f_1 = 20$, $f_0 = 4$, $f_2 = 11$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{16}{25} \times 10 = 76.4$ (1).	1+1
12	Take $a = 115$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2; $\sum f_i = 50$ and $\sum f_i u_i = -5$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 115 + 10 \times \frac{-5}{50} = 114$ (1).	1+1
13	Cumulative frequencies: 14, 25, 31, 51, 65, 83, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 100–110: $l = 100$, $cf = 31$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 100 + \frac{50-31}{20} \times 10 = 109.5$ (1).	1+1
14	Highest frequency is 21, so the modal class is 110–120; $l = 110$, $f_1 = 21$, $f_0 = 20$, $f_2 = 18$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 110 + \frac{1}{4} \times 10 = 112.5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Take $a = 35$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2 (1); $\sum f_i = 100$ and $\sum f_i u_i = -30$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 35 + 10 \times \frac{-30}{100} = 32$ (1).	1+1+1
16	Cumulative frequencies: 11, 30, 37, 57, 70, 89, 100; $N = 100$, so $\frac{N}{2} = 50$ (1); Median class 50–60: $l = 50$, $cf = 37$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{50-37}{20} \times 10 = 56.5$ (1).	1+1+1
17	$\sum f_i = 100$ and $\sum f_i x_i = 7290$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7290}{100} = 72.9$ (1); $l = 50$, $f_1 = 21$, $f_0 = 17$, $f_2 = 9$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{4}{16} \times 10 = 52.5$ (1).	1+1+1
	OR: Take $a = 95$, $h = 10$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 50$ and $\sum f_i u_i = -1$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 95 + 10 \times \frac{-1}{50} = 94.8$ (1).	1+1+1
18	Cumulative frequencies: 10, 16, 31, 43, 48, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 20–30: $l = 20$, $cf = 16$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{25-16}{15} \times 10 = 26$ (1).	1+1+1
	OR: $\sum f_i = 50$ and $\sum f_i x_i = 2710$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2710}{50} = 54.2$ (1); $l = 50$, $f_1 = 16$, $f_0 = 8$, $f_2 = 8$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{8}{16} \times 10 = 55$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i = 80 + p$ and $\sum f_i x_i = 4460 + 55p$ (2); $\frac{4460 + 55p}{80 + p} = 55.6$ gives $(55 - 55.6)p = -12$ (2); $p = 20$ (1).	2+2+1

	OR: Median class is 90–100 with $f = 10$, and the cumulative frequency before it is $cf = 28 + x$ (2); $94 = 90 + \frac{50-(28+x)}{10} \times 10$ gives $x = 18$ (2); $x = 18$; then $y = 100 - 60 - x = 22$ (1).	2+2+1
20	$\sum f_i = 35 + p$ and $\sum f_i x_i = 895 + 25p$ (2); $\frac{895+25p}{35+p} = 25.4$ gives $(25 - 25.4)p = -6$ (2); $p = 15$ (1).	2+2+1
	OR: Median class is 60–70 with $f = 20$, and the cumulative frequency before it is $cf = 38 + x$ (2); $64 = 60 + \frac{50-(38+x)}{20} \times 10$ gives $x = 4$ (2); $x = 4$; then $y = 100 - 75 - x = 21$ (1).	2+2+1

Mistakes to avoid:

- Taking cf of the median class as its own cumulative frequency; cf in the formula is that of the class *before* the median class.
- Choosing the median class by comparing with n instead of $\frac{n}{2}$.
- Confusing f_0 and f_2 (preceding and succeeding class frequencies) in the mode formula.
- Using the class limits instead of the class marks x_i in $\sum f_i x_i$.
- In the step-deviation method, forgetting to multiply $\frac{\sum f_i u_i}{\sum f_i}$ by h.
- Using the modal class (highest frequency) as the median class, or vice versa.

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